Thomas Szymanski

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EDUCATION

NYU Steinhardt School of Culture, Education, and Human Development Expected May 2027
Master of Science in Applied Statistics for Social Science Research; Concentration: General Applied Statistics GPA: 3.9

Macaulay Honors College at Hunter College June 2025
Bachelor of Arts; Major: Mathematics; History | Dean's List GPA: 3.8

RELEVANT COURSEWORK

NYU Steinhardt (Graduate): Multi-Level Modeling, Messy Data and Machine Learning, Causal Inference, Modern Approaches in Measurement, Generative AI for the Social Sciences, Databases and Data Science, Generalized Linear Models and Extensions, Supervised and Unsupervised Machine Learning, Statistical Practice and Proficiency, Statistical Computing, Frequentist Inference
Hunter College (Undergraduate): Advanced Probability Theory I and Geometries I (Graduate Level), Mathematical Analysis I, Abstract Algebra, Linear Algebra, Differential Equations, Vector Analysis

EXPERIENCE

President, **Society for Statistics Students**, NYU Steinhardt, New York, NY September 2026 – Present
- Co-founded the Portfolio Projects Initiative (nyu-statistics-club.vercel.app/portfolio-projects), an independent, student-run research program pairing MSc students with partner organizations on projects in business analytics, data journalism, and AI.
- Recruited student applicants, organized the research timeline and GitHub publishing model, and currently lead a research team.

Independent Study Researcher, **NYU Steinhardt**, New York, NY May 2026 – July 2026
- Conducted supervised research under Prof. Klint Kanopka, importing Generalizability Theory (G-theory) as a replicable model evaluation tool to determine which experimental conditions explain the performance and failure types of an AI agentic system.
- Successfully engineered an experimental design for analyzing LLM task completion on an established web search benchmark, establishing a standardized evaluation pipeline for other models and agentic harnesses.
- Proved that task structure and workflow hierarchy impact agentic reliability far more than model reasoning levels.

ARAC Tutor, **America Reads and Counts at NYU**, New York, NY October 2025 – Present
- Increased Mathematics and English scores in a 5th-grade classroom by providing tutoring and developing original materials.
- Led targeted interventions for English learners, leveraging Spanish and Russian proficiency while acquiring functional Mandarin on the job to communicate instructions and bridge language gaps.
- Applied graduate-level statistics to analyze student assessment data, identifying patterns of need and informing differentiated instruction.

Student Practitioner, **Consulate-General of the Republic of Poland**, New York, NY May 2024 – June 2024
- Organized research on Polish business activities into Excel reports and databases for consular staff.
- Synthesized legal documents in Polish for consular proceedings.
- Supported Polish-language citizenship and documentation interviews, guiding diaspora applicants in Ukrainian and Russian.

SELECTED PROJECTS

LLM & Machine Learning Evaluation of NYS ELA Reading Passages *Spring 2026*
Collaborators: Howell Lu, Ruiting Shen — github.com/tfs6492/nys-ela-llm-eval
- Tested whether ELA State Test passages matched their assigned grade levels by benchmarking LLMs against machine learning.
- Engineered LLM prompts and API workflows while translating design decisions for teammates without AI training.
- Found linguistic features do not track grade level, with all model outputs showing systematic deviation from assigned grade level.

Psychometric Modeling of Duolingo Spanish Reverse-Tap Exercises *Spring 2026*
Collaborator: Shen Shen Lau — github.com/tfs6492/duolingo-es-psychometrics
- Investigated a Duolingo dataset to detect underlying user profiles and what linguistic features cause learners the most difficulty.
- Coded models in R that revealed core patterns in the dataset and uncovered hidden user traits.
- Pivoted to analyzing user response times to address severe data sparsity that violated traditional modeling assumptions.
- Successfully identified verb tense/mood as the single driver of both item difficulty and response time.

Social Connectedness and Central American Migration to the U.S. *Spring 2026*
Collaborator: Genie Kwak — github.com/tfs6492/sci-central-america-migration
- Verified Meta's Social Connectedness Index (SCI) as a proxy tool to study migration flows.
- Joined and cleaned DHS permanent resident data using R (Tidyverse) to cross-reference connectivity with migration flows.
- Communicated findings through a written report and data visualizations generated with R (ggplot2).
- Found Central American nations' connectivity level with the U.S. far exceeds the global average, reflecting share in migration.

SKILLS

Programming Languages: R (advanced), SQL (proficient), Python (beginner)
Development Environment: Git, CLI, RStudio, Zed, Claude Code, Google Antigravity, Hermes Agent, Pi Agent
Local LLM Stack: Unsloth, Ollama, llama.cpp
Foreign Languages: Polish (fluent), Spanish, Russian (advanced), German, Ukrainian (lower-intermediate), Mandarin (beginner)